AspireAI

ASPIRE AI

AI-Powered Career Guidance System for Sri Lankan Students

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OVERVIEW

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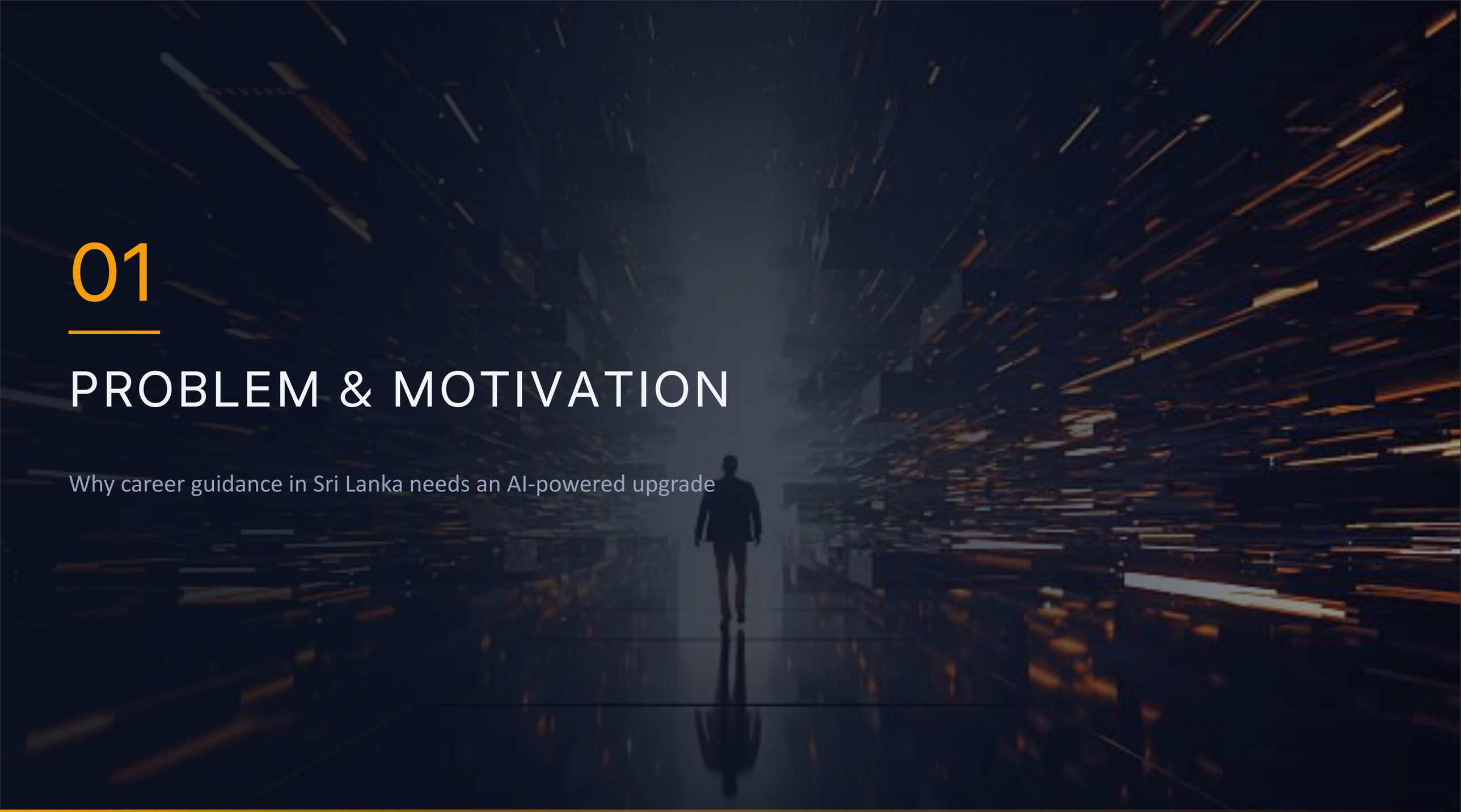


01

PROBLEM & MOTIVATION

Why career guidance in Sri Lanka needs an AI-powered upgrade

Why Career Guidance Needs an Upgrade

1 ROOT CAUSE

The Core Problem

Students choose A/L streams and careers with limited guidance, influenced by social pressure rather than skills and interests.

2 KEY ISSUES

- Poor A/L stream selection
- Higher dropout risk
- Workforce skill mismatch
- Limited non-traditional guidance

Impact Flow

Limited Guidance / Social influence overrides personal skills & interests

interests

↓

Misaligned Choices / Wrong streams, courses, and career paths selected

selected

↓

Dropout & Skill Mismatch / Higher education dropout + workforce gaps

gaps

This cycle perpetuates without intelligent intervention

02

SOLUTION OVERVIEW

An intelligent platform for personalized academic and career recommendations

AI-Powered Career Guidance System

"Personalized academic + career recommendations aligned with skills, interests, and available pathways"

INPUTS

- Academics
- Interests
- Skills

↓

AI ENGINE

Intelligent Processing

ML Prediction + RAG Context + LLM Reasoning

↓

OUTPUTS

- Study Streams
- Career Paths

CORE FEATURES

✓ Predict suitable A/L streams for O/L students

✓ Recommend university courses for A/L students

✓ Evaluate soft skills & communication abilities

✓ Learn-to-earn pathways for non-traditional learners

KEY TECHNOLOGIES

ML Models · RAG · LLMs

Four Intelligent Modules

Stream Guidance System

Predicts suitable A/L streams for O/L students using Machine Learning models

Soft Skill Development System

Evaluates communication abilities and soft skills through AI-powered assessment

AI Guidance Platform

University Course Prediction

Recommends university courses based on A/L performance and interests

Learn-to-Earn Pathway Generator

Creates alternative career pathways for non-traditional learners learners

03

SYSTEM ARCHITECTURE

Tech stack, models, and end-to-end pipeline design

Tech Stack & Pipeline

FRONTEND
Next.js

BACKEND
Node.js / FastAPI

AI LAYER
ML + RAG + LLMs

DATABASE
MongoDB

END-TO-END PIPELINE

Students → Input → API Gateway → AI MODULES → AI/ML Layer

AI MODULES
- Stream & Course Suggestion
- University Recommender
- Pathway Engine
- Soft Skill Enhancement
- Recommender

System Models

Intelligent Stream Guidance

ML-based A/L stream prediction for O/L students using academic history academic history and interests

University Course Prediction

Course recommendation engine matching A/L performance with with university programs

Learn-to-Earn Diagram Generator

Personalized pathway creation for non-traditional learners with skill-gap analysis

Soft Skill Development Module

Communication Skill Model + Problem Solving Model for holistic holistic evaluation

AI/ML Technologies Used

- XGBoost / Random Forest
- ARIMA / Prophet
- BERT / NLP Models
- CatBoost / KNN
- Regression Models
- Agentic RAG
- LLM Integration
- Vector Database
- Context Retrieval

04

USER JOURNEYS

How different learner types interact with the system

Who the System Serves

O/L Student

O/L results + interests → AI Analysis → Suggested A/L streams

Inputs academic results and personal interests. Receives AI-powered analysis with personalized A/L stream recommendations aligned to strengths and goals.

A/L Student

A/L performance → Course Matching → University courses

Submits A/L performance data. System matches scores with university course university course requirements, recommending optimal degree programs and programs and institutions.

Soft Skill Seeker

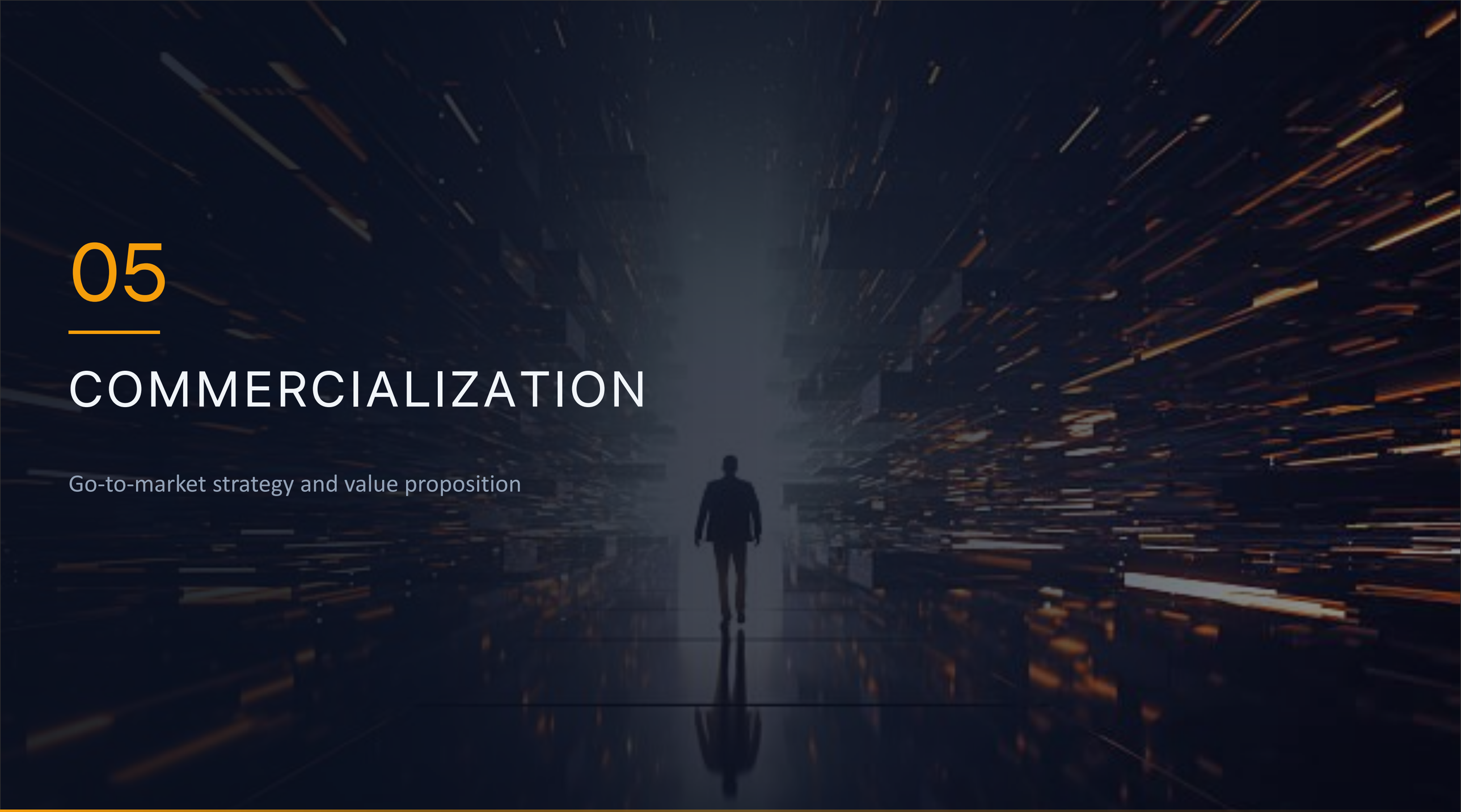
Current skill level → AI Evaluation → Development plan

Provides current skill assessments. Receives comprehensive evaluation of evaluation of communication and problem-solving abilities with targeted targeted improvement plans.

Non-traditional Learner

Goals + skills → Gap Analysis → Learn-to-earn pathway

Shares career goals and existing skills. System generates personalized learn-to-learn-to-earn pathways with alternative qualifications and skill-building routes. building routes.



05

COMMERCIALIZATION

Go-to-market strategy and value proposition

Go-to-Market & Value Proposition

Students

- O/L students seeking guidance
- A/L students applying for uni
- Personalized recommendations

Parents

- Clearer pathway understanding
- Data-driven decisions
- Reduced anxiety about choices

Institutions

- Schools & career centers
- Reduced counseling workload
- Government workforce planning

Value Proposition Summary

01

Personalized Guidance

AI-tailored recommendations based on individual skills and interests

02

Data-Driven Decisions

Replace social pressure with evidence-based career pathways

03

Inclusive Access

Pathways for both traditional students and non-traditional learners

06

REVENUE MODEL

Sustainable monetization through freemium and institutional partnerships

partnerships

Freemium + Institutional Plans

Freemium

For Students

- ✓ Basic career guidance
- ✓ Stream suggestions
- ✓ General pathway info
- ✓ Free for all students

POPULAR

Premium

Advanced Features

- ✓ Advanced analytics
- ✓ Detailed career reports
- ✓ Skill assessments
- ✓ Priority support

Institutional

Subscriptions

- ✓ School-wide access
- ✓ University partnerships
- ✓ Admin dashboard
- ✓ Bulk licensing

THANK YOU

Questions?

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AI-Powered Career Guidance System